
Routing Visual Evidence Under Domain Shift: Three-Way Factorization for Cross-Domain Audio-Visual Deepfake Detection

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Abstract

Multimodal models are expected to integrate complementary signals across modalities, yet in practice they often rely on spurious shortcuts from a single modality. This failure is particularly evident in cross-domain audio-visual deepfake detection, where models may ignore visual manipulation evidence and instead rely on easier but unreliable audio cues. A common approach to mitigate such failures is to enforce invariance through regularization or domain-adversarial objectives. However, it remains unclear whether these methods change how models make decisions, or merely reshape representations without altering their usage. In this work, we show that multimodal robustness is not primarily a problem of enforcing invariance, but of structuring how information is used at inference time. We demonstrate that loss-level interventions fail to resolve representation-level entanglement in standard binary decompositions, where modality-specific signals and domain nuisance remain mixed. We introduce a simple three-way factorization that separates shared content, modality-specific task-relevant components, and residual domain-dominant signals. This structural change leads to a qualitatively different behavior: performance gains emerge only in diagnostic settings that require reliance on the correct modality, and do not appear when applying the same objectives without factorization. Crucially, we provide multiple lines of evidence that these improvements arise from a change in inference-time routing rather than representation invariance. Head ablation, input-level masking, and subspace probing show that predictions depend critically on the visual-unique component, while domain information remains present in the representation. In cross-domain audio-visual deepfake detection, this results in substantial improvements on the FV-RA setting, where correct predictions require reliance on visual evidence despite misleading audio signals. Together, our results suggest a shift in perspective: instead of suppressing spurious features, multimodal robustness can be achieved by explicitly structuring the pathways through which decisions are made.

1 Introduction

Multimodal learning aims to combine complementary information from different sources such as vision and audio. In principle, this enables models to make robust decisions even when one modality is unreliable. In practice, however, multimodal models often rely on shortcut signals from a single modality, failing to integrate evidence across modalities. This issue is particularly pronounced in cross-domain audio-visual deepfake detection, where models may ignore visual manipulation evidence and instead rely on easier but unreliable audio cues. A common approach to mitigating such failures is to enforce invariance, for example through domain-adversarial training or feature

regularization. These methods aim to remove spurious correlations from learned representations, under the assumption that robustness follows from eliminating nuisance information. While such approaches can improve performance under certain conditions, it remains unclear whether they actually change how models make decisions, or simply reshape representations without affecting how they are used at inference time.

This raises a fundamental question: do existing methods improve robustness by learning better representations, or by changing how models use those representations? In this work, we show that in standard binary decompositions, loss-level interventions are insufficient to resolve this issue. Even with domain-adversarial objectives, modality-specific task signals and domain-related nuisance remain entangled within the same branch. As a result, the model lacks a structural mechanism to separate useful modality evidence from shortcut signals, leading to unstable behavior under domain shift. We argue that the problem is therefore not primarily one of objective design, but of representational structure. To address this, we introduce a simple three-way factorization that decomposes multimodal representations into (1) shared content, (2) modality-specific task-relevant components, and (3) residual domain-dominant signals. By explicitly separating these roles and restricting the prediction head to use only task-relevant components, the model is encouraged to rely on the correct modality.

Our empirical results reveal a consistent pattern. Performance improvements are concentrated in diagnostic settings that require reliance on the correct modality, while other categories remain near ceiling. Controlled comparisons show that these gains cannot be explained by additional objectives alone, and only emerge when the representation is factorized. More importantly, we provide direct evidence that the effect is driven by a change in inference-time routing. Head ablation and input-level masking demonstrate that predictions depend critically on the visual-unique component, whereas non-factorized models remain largely insensitive. Notably, domain information remains present in the representation, indicating that robustness is achieved without enforcing full invariance. In the context of cross-domain audio-visual deepfake detection, these effects are most clearly observed in the FV-RA setting, where audio shortcuts are uninformative and correct predictions require reliance on visual manipulation evidence. The proposed factorization substantially improves performance in this diagnostic regime. These findings suggest a different perspective on multimodal robustness. Rather than attempting to remove spurious signals from representations, it may be more effective to structure representations such that models are encouraged to route decisions through the appropriate modality. We summarize our contributions as follows:

- We show that loss-level approaches fail to disentangle modality-specific and domain signals in standard binary decompositions, leading to shortcut reliance in cross-domain audio-visual deepfake detection.
- We introduce a simple three-way factorization that provides a dedicated pathway for modality-specific task-relevant information, enabling more reliable use of visual manipulation evidence.
- We provide multiple lines of evidence that improvements arise from changes in inference-time routing rather than representation invariance, validated through ablation, masking, and subspace analysis.
- We demonstrate substantial improvements on cross-domain audio-visual deepfake detection, particularly in the FV-RA setting, where correct predictions require reliance on visual evidence despite misleading audio signals.

2 Related Work

Audio-visual deepfake detection. Multimodal deepfake detection leverages the fact that realistic speech, lip motion, and facial dynamics must stay mutually consistent over time. Early work explored direct audio-visual fusion and synchronization cues [? ?]. More recent methods use self-supervised anomaly detection, speech representation learning, or explicit combinations of modality-shared semantic mismatch and modality-specific forensic traces [? ? ? ?]. These approaches are more robust than purely artifact-based detectors, but most still optimize a single prediction pathway and therefore do not directly test which subspace the detector routes its decisions through under shift.

Evaluation protocols. Most audio-visual deepfake work still reports in-dataset or within-benchmark splits. FakeAVCeleb-based evaluation commonly uses train/test splits and leave-one-manipulation protocols [? ?], while AV-Deepfake1M reports official in-dataset benchmarks on AV1M, LAV-DF, and FAVC [?]. The closest prior cross-dataset bridge is AVH-Align, which reports supervised AV1M→FAVC scores while diagnosing leading-silence shortcuts [?]. Our contribution is therefore not the existence of FakeAVCeleb as a target dataset, but an AV1M→FAVC cross-domain stress test with modality-category reporting and routing analyses focused on FV-RA.

Speech-aware multimodal representations. Large-scale audio-visual speech models, such as AV-HuBERT [?], learn strong multimodal features that capture both local synchronization and longer-term speech structure. Such representations have proven useful beyond recognition, including forensic analysis and synchronization-based detection [? ?]. Our method uses the same intuition, but adds an explicit factorization layer so that the downstream detector can discard nuisance factors instead of letting them leak into the prediction head.

Multimodal factorization. Generic multimodal representation learning has explored decomposing features into shared/synergistic, unique, and redundant components. The closest such prior is FCD [?], a plug-and-play factorization for general multimodal prediction that reserves synergistic and unique factors while removing redundant, task-agnostic uncertainty. TRIROUTE uses a related three-part vocabulary, but the residual factor has the opposite operational role. In FCD-style decompositions, the residual or redundant component is primarily unwanted uncertainty to suppress. In TRIROUTE, the residual is an intentionally supervised *information sink*: it is trained to retain domain-predictive shortcut evidence, is structurally excluded from the fake classifier, and is paired with adversarial pressure only on the task branch. The resulting design is therefore not a generic decomposition module, but a domain-routing mechanism: shortcut evidence is attracted into Z_{res} while the prediction path is forced to rely on shared content and modality-unique manipulation cues.

This design is related to, but distinct from, earlier shared-private and debiasing lines. Domain Separation Networks (DSN) explicitly split shared and private domain components for adaptation [?], and DIVA learns separate latent variables for domain, class, and residual variation in a generative framework [?]. Bias-adversarial approaches such as Learning Not to Learn [?] and Learning from Failure [?] suppress shortcut exploitation through adversarial or dual-network training. Our method differs in the combination of three properties used together for routing control: (i) a supervised domain-predictive residual sink, (ii) structural exclusion of that sink from the fake head, and (iii) explicit routing validation under cross-domain FV-RA stress tests.

Domain alignment and adversarial adaptation. Classical unsupervised domain adaptation often tries to make source and target features globally aligned. DANN [?] trains a gradient-reversal domain classifier on top of the feature representation so that the encoder learns features from which the domain is difficult to predict. CORAL-style alignment instead matches second-order feature statistics across domains, either as a shallow covariance transform or as a differentiable deep loss [?]. Both families are useful generic adaptation tools, but in cross-domain audio-visual forensics their pressure can be misdirected when applied to an undifferentiated representation: domain-correlated nuisance, speech content, and manipulation evidence may be entangled, so global alignment can suppress useful forensic signal or leave the final classifier free to route through remaining shortcuts. Our design changes the question from “can all features be aligned?” to “which features is the detector allowed to use?” We separate a dedicated residual branch whose role is to absorb domain-correlated signal, pair it with a forward domain-prediction objective ($\mathcal{L}_{\text{dom}}$), and apply adversarial pressure only to the task branch. In this sense, TRIROUTE is complementary to DANN/CORAL: it does not merely align all features, but changes the admissible prediction subspace.

3 Problem Setting

We study cross-dataset audio-visual deepfake detection. Let $\mathcal{D} = \{(x_i^v, x_i^a, y_i, d_i)\}_{i=1}^N$ denote training samples where x_i^v is a sequence of visual observations, x_i^a is the synchronized audio, $y_i \in \{0, 1\}$ is the video-level real/fake label, and $d_i \in \{1, \dots, M\}$ indicates a nuisance partition available during training. Training uses AV-Deepfake1M clips for task supervision. In the latest reported setting, nuisance supervision is provided only by source-side speaker-group pseudo-labels

derived from AV1M ($d = 0$ or $d = 1$ by speaker-ID hash), so no FakeAVCeleb data is used at any stage of training, validation, checkpoint selection, or hyperparameter selection.

Our goal is to learn a scoring function $f(x^v, x^a)$ that remains predictive when the nuisance statistics associated with d shift at test time. We therefore seek a representation in which manipulation evidence remains discriminative for y , while predictable but non-transferable signals are isolated rather than implicitly used by the classifier. We do not claim that the learned task branch becomes fully domain-invariant; the claim we test is weaker and more mechanism-focused: factorization changes representation usage and downstream routing under shift.

4 Method

4.1 Overview

Figure ?? gives a conceptual overview of TRIROUTE. Audio-visual forensics under dataset shift requires separating exactly three qualitatively distinct signal types: cross-modal speech content that should generalize across datasets, modality-unique manipulation evidence that is the detection target, and domain-correlated nuisance that must not reach the prediction head. We therefore design a three-branch factorization in which each branch has an explicit semantic role and a corresponding supervision signal, and the prediction head is structurally restricted to the first two. Starting from frozen AV-HuBERT features $z_v, z_a \in \mathbb{R}^{1024}$, we first extract shared cross-modal factors and modality-specific hidden states:

$$[s_v, h_v] = F_v(z_v), \quad [s_a, h_a] = F_a(z_a), \quad (1)$$

where s_v and s_a denote shared task-relevant factors and h_v, h_a collect the remaining modality-specific signal. We then split the modality-specific states into unique task-relevant and residual components:

$$[u_v, r_v] = G_v(h_v), \quad [u_a, r_a] = G_a(h_a). \quad (2)$$

Here u_v and u_a are modality-unique task factors, while r_v and r_a form the residual branch that is allowed to absorb shortcut and domain signal. The task representation and residual representation are

$$Z_{\text{task}} = \text{cat}(s_v, u_v, s_a, u_a), \quad Z_{\text{res}} = \text{cat}(r_v, r_a). \quad (3)$$

For readability, we refer to the shared factor as *content*, the visual-unique task factor as the key *manipulation* carrier for FV-RA, and the residual branch as the *domain* factor.

4.2 Three-Way Factorization

Manipulation factor. The manipulation factor is intended to encode modality-unique evidence that cannot be explained by authentic cross-modal correspondence. In practice, the visual-unique component u_v is the most important carrier for the visual-fake category FV-RA, while u_a supports audio-fake cases. These unique components feed the task head together with the shared factors.

Content factor. The content factor captures shared speech semantics and identity-preserving motion that should remain useful across domains. We align the shared factors across modalities using an audio-visual alignment objective:

$$\mathcal{L}_{\text{align}} = - \sum_{t=1}^T \log \frac{\exp(\text{sim}(s_{v,t}, s_{a,t})/\tau)}{\sum_{j \in \mathcal{N}(t)} \exp(\text{sim}(s_{v,t}, s_{a,j})/\tau)}, \quad (4)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity, τ is a temperature, and $\mathcal{N}(t)$ is a local temporal window.

Domain factor. The residual branch is explicitly optimized to predict the source domain:

$$\mathcal{L}_{\text{dom}} = \text{CE}(C_d(Z_{\text{res}}), d). \quad (5)$$

This branch acts as an information sink for codec, resolution, acquisition, and dataset-construction artifacts.

176 4.3 Detection and Domain Push-Pull Objectives

177 We predict a clip-level fake probability from the task representation:

$$\hat{y} = C_y(Z_{\text{task}}), \quad (6)$$

178 and optimize binary cross-entropy:

$$\mathcal{L}_{\text{fake}} = \text{BCE}(\hat{y}, y). \quad (7)$$

179 To keep task-relevant and residual factors separated, we apply an orthogonality penalty between
180 them:

$$\mathcal{L}_{\text{ortho}} = \|s_v^\top r_v\|_F^2 + \|s_a^\top r_a\|_F^2 + \|u_v^\top r_v\|_F^2 + \|u_a^\top r_a\|_F^2. \quad (8)$$

181 To suppress easy domain prediction from the task branch, we apply a gradient-reversal domain
182 classifier:

$$\mathcal{L}_{\text{adv}} = \text{CE}(C_{\text{adv}}(\text{GRL}(Z_{\text{task}})), d), \quad (9)$$

183 while $\mathcal{L}_{\text{dom}}$ simultaneously attracts domain information into Z_{res} . This push-pull effect does not aim
184 to make Z_{task} fully domain-invariant; instead it makes shortcut routing into the prediction head less
185 attractive.

186 The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{fake}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{dom}} \mathcal{L}_{\text{dom}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{ortho}} \mathcal{L}_{\text{ortho}}. \quad (10)$$

187 **Loss-weight selection.** The four weights in Eq. (??) control distinct roles rather than interchange-
188 able regularizers: λ_{align} stabilizes shared audio-visual content, λ_{dom} attracts nuisance information
189 into Z_{res} , λ_{adv} discourages easy domain prediction from Z_{task} , and λ_{ortho} keeps the task and residual
190 factors separated. We do not run a target-domain grid search and do not use FakeAVCeleb labels,
191 category AUCs, or test-set statistics when setting these weights. The values are fixed once using
192 source-only validation diagnostics: auxiliary losses are kept on a comparable scale to the fake-
193 classification loss, the residual branch must remain domain-predictive on held-out source partitions,
194 and the task branch must retain source-validation AUC. The same fixed values are then used for
195 all reported AV1M→FAVC variants. This avoids a high-dimensional target-driven hyperparameter
196 search and keeps the main comparison focused on representation structure rather than per-setting
197 weight tuning.

198 4.4 Why the Factorization Helps

199 The key design choice is that the detector does not consume the entire multimodal representation.
200 Instead, it reads only the shared and unique task factors and excludes the residual branch from the
201 prediction head. This matters because many dataset shortcuts are genuinely predictive on the source
202 domains; without an explicit nuisance branch, the model has no incentive to avoid them. By exposing
203 a visual-unique task channel and a separate residual/domain channel, TRIROUTE turns better evidence
204 usage into an optimization target rather than a hope.

205 5 Experiments

206 5.1 Datasets and Evaluation

207 We train on AV-Deepfake1M (AV1M) [?], a large-scale dataset of over one million clips spanning
208 diverse synthesis pipelines, and evaluate on the FakeAVCeleb (FAVC) [?] 70/30 public test
209 split ($N = 6,667$), following the evaluation protocol of recent shortcut analyses of audio-visual
210 deepfake datasets [?]. We report threshold-free AUC and average precision (AP) separately for three
211 informative categories: (i) RealVideo-FakeAudio (RV-FA), (ii) FakeVideo-RealAudio (FV-RA), and
212 (iii) FakeVideo-FakeAudio (FV-FA). FV-RA is the most diagnostic category because audio shortcuts
213 are intentionally unhelpful there, forcing the model to use genuine visual manipulation evidence. We
214 report all categories for completeness, but center the analysis on FV-RA because RV-FA and FV-FA
215 are near ceiling in this benchmark and are therefore less informative about cross-domain visual
216 routing. We use AUC as the primary comparison metric: AP is also reported for completeness, but it

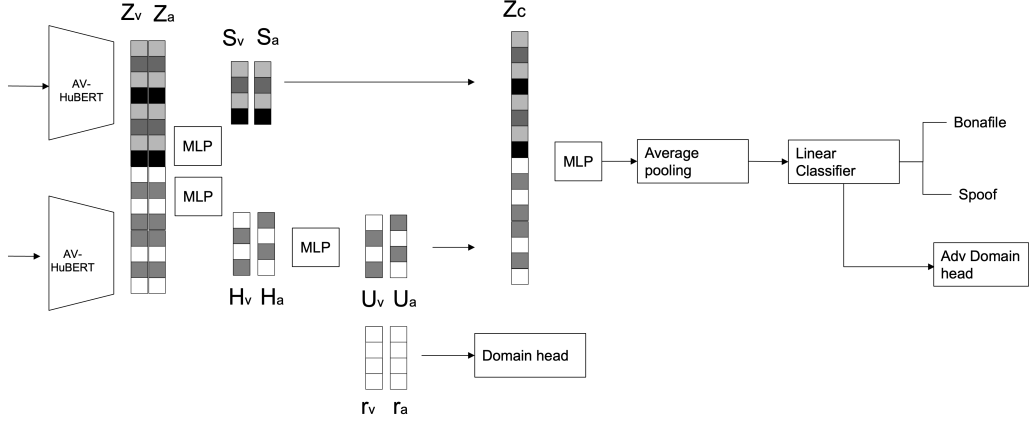


Figure 1: **NEED TO REDRAW** Conceptual overview of TRIROUTE. Frozen AV-HuBERT features are decomposed into shared cross-modal factors (s_v, s_a), modality-unique factors (u_v, u_a), and residual factors (r_v, r_a). The task head reads only the task-relevant shared and unique factors, while domain push-pull discourages easy domain prediction from Z_{task} and concentrates nuisance information in Z_{res} . We use the term “routing” to refer to which of these factors the final detector actually relies on at inference time.

217 is high across many FAVC category splits because the one-vs-rest evaluation is strongly imbalanced,
 218 so AP is less sensitive than AUC to the ranking failures studied here. We leave threshold-dependent
 219 accuracy out of the main comparison because cross-dataset operating thresholds are not calibrated in
 220 this protocol.

221 5.2 Baselines

222 We compare TRIROUTE against AVH-Align/sup [?], the only published method that trains on AVIM
 223 and evaluates on FAVC under a directly compatible AV-HuBERT protocol, and a progression of
 224 internal baselines that isolate each design choice. AVFF [?] trains primarily on FakeAVCeleb (reversing
 225 the transfer direction), precluding a direct AVIM→FAVC comparison under the source-trained
 226 protocol. The **two-way baseline** applies a coarser two-branch factorization without domain routing.
 227 The **two-way + DANN/GRL control** is the DANN-style baseline in our protocol: a gradient-reversal
 228 domain classifier is applied to the two-way representation without a residual shortcut sink or three-
 229 way factorization. The **two-way + CORAL control** applies the corresponding statistic-matching
 230 baseline, aligning source and target covariance statistics on the same two-way representation. Both
 231 are global domain-adaptation controls rather than routing mechanisms. The **three-way factorization**
 232 **without domain push-pull** applies the same three-branch structural separation as TRIROUTE but
 233 omits the explicit domain-prediction objective on the residual and the adversarial objective on the
 234 task branch. This row is the closest control to an FCD-style decomposition in our setting: it keeps
 235 the shared/unique/residual split and excludes the residual from the task head, but does not make the
 236 residual branch a supervised domain sink. All variants use identical AV-HuBERT features and the
 237 same classifier backbone.

238 5.3 Implementation Details

239 TRIROUTE is built on frozen AV-HuBERT [?] features ($C = 1,024$, 25 fps, mouth-crop visual
 240 stream). The factorization modules and classifier are lightweight MLPs (three to four layers, LayerNorm, ReLU). The shared, unique, and residual components use 256 dimensions per modality.
 241 Training uses Adam with learning rate 10^{-3} , batch size 1 (variable-length clips), and a maximum of
 242 50 epochs with patience-20 early stopping on source-validation AUC. All checkpoint selection and
 243 hyperparameter choices use only labelled source-domain validation data; target-domain labels are
 244 used only once for final reporting. We fix $\lambda_{\text{align}} = 0.5$, $\lambda_{\text{dom}} = 1.0$, $\lambda_{\text{adv}} = 1.0$, and $\lambda_{\text{ortho}} = 0.1$,
 245 with GRL $\alpha = 1.0$.
 246

For the latest result set, nuisance labels are derived entirely from the AV1M training set via a speaker-group hash ($d = \text{hash}(\text{speaker_id}) \bmod 2$), assigning all clips of a speaker to the same binary group; no FakeAVCeleb data enters training, validation, or model selection. This is deliberately weaker than true target-domain supervision and should be understood as a source-side nuisance partition rather than an oracle domain label. Unless otherwise stated, quantitative results report the mean over three independent seeds.

For fairness, the DANN/GRL and CORAL controls use matched optimization settings (Adam 10^{-3} , GRL $\alpha=1.0$, CORAL weight $\lambda=1.0$) under the same seed/checkpoint policy. We do not perform target-aware tuning sweeps on FakeAVCeleb labels; $\alpha=1.0$ is used as a standard default for binary domain discrimination in our protocol.

Seed policy. All model families in the main ablation table are reported as means over three independent seeds under the same source-validation checkpointing rule. We do not select checkpoints or seeds using FakeAVCeleb labels. The purpose of the seed sweep is to verify that the ordering of the main variants is stable, not to tune the method per target category.

6 Results and Analysis

6.1 Cross-Domain Detection Performance

Why FV-RA is the diagnostic setting. We treat FV-RA (FakeVideo-RealAudio) as the central evaluation category throughout this section because it is the only category in which the audio stream cannot explain the fake label: the audio is genuine, so any correct “fake” decision must rely on visual manipulation evidence. RV-FA and FV-FA, in contrast, both contain manipulated audio and are routinely solved by audio-driven cues alone – consistent with the near-ceiling AUCs observed for every model in Table ?? . FV-RA therefore forces the detector to use the correct modality under conflicting modalities, and a model’s FV-RA AUC is what reveals whether it is actually routing through visual evidence or merely riding an audio shortcut. For this reason we organize both narrative and figures around FV-RA and treat the saturated audio-fake categories as background.

Table ?? compares TriROUTE against the AV1M-trained AVH-Align/sup checkpoint from ?] and a progression of internal ablation baselines. AVH-Align/sup is a strong two-branch supervised baseline that achieves FV-RA AUC of 0.519 despite training on the same source data, establishing a competitive non-factorized ceiling. The two-way baseline achieves near-ceiling performance on RV-FA and FV-FA but falls below chance on FV-RA (0.437 AUC). Because RV-FA and FV-FA are already saturated under this protocol, we do not interpret the overall gain as uniform improvement across all manipulation categories; the non-saturated test is FV-RA, where the audio stream is real and therefore cannot explain the fake label. The two global adaptation controls show marginal improvement over the two-way baseline: DANN/GRL reaches 0.496/0.955 FV-RA and CORAL reaches 0.521/0.956, but both remain near or below chance. This confirms that global domain alignment alone is insufficient when no residual shortcut sink is available: reducing domain discrepancy does not specify which evidence the classifier should use for FV-RA. Three-way factorization without domain push-pull improves FV-RA to 0.626/0.973, but remains well below the latest TriROUTE result.

Notably, training the two-way baseline to full convergence under the same protocol *lowers* its FV-RA AUC from 0.523 (20 epochs) to 0.437 (100 epochs). This drop is itself diagnostic: with more training, a non-factorized model amplifies its reliance on the audio shortcut for FV-RA decisions. The baseline does not fail because it is undertrained—it fails because the prediction pathway is structurally exposed to shortcut signals. Two further observations rule out the alternative “it is just better regularization” explanation: (i) adding the same domain-adaptation objectives on top of the two-way representation (DANN/GRL and CORAL) does not close the FV-RA gap, with both controls remaining at or below the two-way baseline; (ii) three-way factorization without any domain push-pull objective already lifts FV-RA to 0.626. The same family of objectives therefore fails on a binary representation but becomes effective once the representation is factorized — it is the structural separation, not added optimization or regularization, that turns these objectives into a source of gain.

We report TriROUTE as the mean over three independent seeds. This source-side setting uses only AV1M speaker-group nuisance labels and no FakeAVCeleb data for training, validation, or model selection. Despite this strict protocol, the latest TriROUTE result reaches 0.838 FV-RA AUC, improving over the two-way baseline by +0.401 AUC and over the three-way no-push-pull control by

Table 1: Cross-domain detection on FakeAVCeleb (AV1M $\rightarrow$ FAVC, $N = 6,667$). Each numeric cell reports AUC/AP; all rows report the mean over three independent seeds. AVH-Align/sup is the single published AV1M-trained checkpoint from [?]. RV-FA = RealVideo-FakeAudio; FV-RA = FakeVideo-RealAudio; FV-FA = FakeVideo-FakeAudio. The latest TRIROUTE row uses AV1M speaker-group nuisance labels only; no FakeAVCeleb data enters training, validation, or model selection.

Method	Structure	Overall	RV-FA	FV-RA (key)	FV-FA
AVH-Align/sup [?]	two-way supervised	0.777/0.994	0.999/0.999	0.519/0.955	0.999/1.000
Two-way baseline	two-way factorization	0.739/0.993	1.000/1.000	0.437/0.949	0.998/1.000
Two-way + DANN/GRL control	two-way + GRL	0.767/0.994	1.000/1.000	0.496/0.955	1.000/1.000
Two-way + CORAL control	two-way + CORAL	0.769/0.994	1.000/1.000	0.521/0.956	0.980/0.999
Three-way, no domain push-pull	three-way factorization	0.827/0.996	1.000/1.000	0.626/0.973	0.999/1.000
TRIROUTE (ours)	three-way + push-pull	0.922/0.998	0.999/0.999	0.838/0.989	0.994/1.000

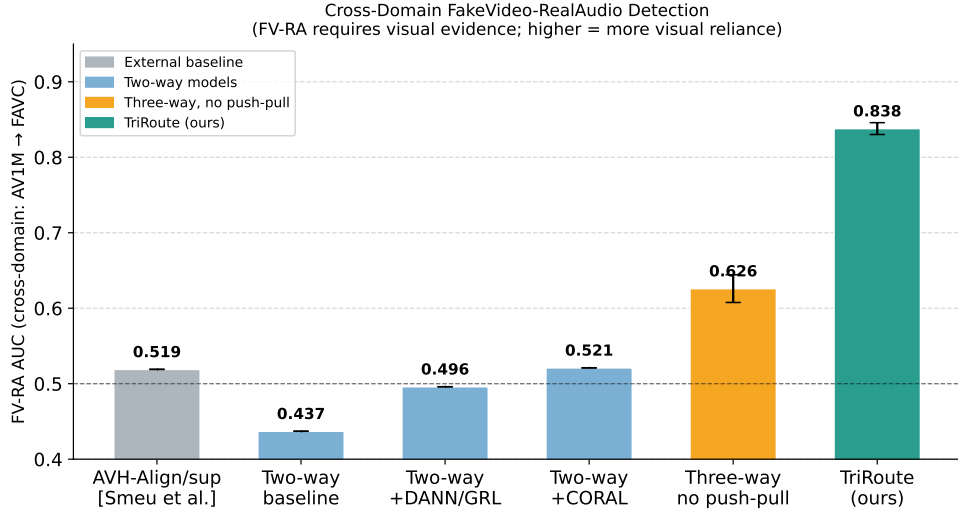


Figure 2: Cross-domain FV-RA AUC on FakeAVCeleb (AV1M $\rightarrow$ FAVC, $N=3,158$ FV-RA clips), mean over three independent seeds with error bars (one standard deviation). Dashed line marks chance (0.5). Bars left to right: AVH-Align/sup [?], our two-way baseline, two global domain-adaptation controls (DANN/GRL $\alpha=1$, CORAL), three-way factorization without domain push-pull, and TRIROUTE. Companion view of the FV-RA column in Table ??: every two-way variant sits at or below chance, while TRIROUTE is the only model that separates cleanly above 0.8. The error bars are tight relative to this gap, indicating the +0.4 AUC improvement on the visual-routing diagnostic is not seed-driven.

300 +0.212 AUC. The overall AUC also rises to 0.922, while RV-FA and FV-FA remain near ceiling. The
301 key observation is therefore not that target-domain data is required, but that robust FV-RA routing
302 emerges once the prediction pathway is structurally separated from the residual shortcut sink.

303 6.2 Mechanism-Level Routing Evidence: Head Ablation

304 **External baseline context.** Table ?? is primarily a mechanism table: its internal baselines isolate
305 whether the gain comes from three-way factorization, global domain alignment, or the residual
306 push-pull design. To make the external comparison explicit, Table ?? separately reports the clos-
307 est published supervised cross-dataset bridge under a compatible AV-HuBERT protocol. Most
308 FakeAVCeleb methods train and test within FakeAVCeleb or report aggregate scores under in-
309 compatible splits, so they do not directly test AV1M $\rightarrow$ FAVC visual routing. The AV1M-trained
310 AVH-Align/sup checkpoint from [?] is therefore the most directly comparable external supervised
311 baseline. It shows the same category-level failure mode as our two-way baseline: RV-FA and FV-FA
312 are near ceiling, while FV-RA remains near chance. TRIROUTE’s improvement is therefore not only
313 relative to our own ablations; it addresses the visual-fake transfer failure of the closest published
314 supervised baseline.

Table 2: Category-level external supervised baseline on FakeAVCeleb (AV1M $\rightarrow$ FAVC). AVH-Align/sup is the AV1M-trained supervised checkpoint from [?], evaluated with the same category protocol as Table ?? . Values are AUC/AP.

Method	Overall	RV-FA	FV-RA	FV-FA
AVH-Align/sup [?]	0.777/0.994	0.999/0.999	0.519/0.955	0.999/1.000
TRIROUTE (ours)	0.922/0.998	0.999/0.999	0.838/0.989	0.994/1.000

Table 3: Head-ablation analysis on FV-RA AUC. Each row zeros out one branch of the trained model and measures the FV-RA drop (Δ FV-RA). Larger drop indicates heavier routing dependence on that branch. Baseline and ablated values should be compared within this intervention protocol rather than directly against Table ?? . All rows report the mean over three independent seeds.

Model	Ablation	Baseline FV-RA	Ablated FV-RA	Δ FV-RA
Two-way baseline	mask visual task branch	0.495	0.461	−0.034
Three-way, no domain push-pull	mask visual branches	0.586	0.473	−0.113
TRIROUTE (ours)	mask manipulation (u_v)	0.836	0.491	− 0.345
TRIROUTE (ours)	mask visual branches	0.836	0.441	− 0.395

Table ?? and Figure ?? quantify how strongly each model’s FV-RA performance depends on its visual branches. For the two-way baseline, zeroing the visual task branch causes a negligible drop of −0.034, confirming that the trained detector barely uses visual evidence for FV-RA decisions. For three-way factorization without domain push-pull, masking visual branches causes a larger drop (−0.113), indicating partial visual usage. In the latest TRIROUTE checkpoints, mean FV-RA falls from 0.836 to 0.491 when u_v is masked and to 0.441 when both visual branches are masked. **The takeaway is direct: only the factorized model is meaningfully sensitive to removing visual components.** The order-of-magnitude gap in $|\Delta$ FV-RA| (−0.034 vs. −0.345) is mechanism-level evidence that TRIROUTE has rerouted FV-RA decisions through the visual-unique factor, while the two-way baseline routes them somewhere else — masking its audio branch instead drops in-domain AUC from 0.997 to 0.528, confirming audio dominance rather than visual capacity failure. This asymmetry is precisely what the three-way factorization is designed to break.

6.3 Qualitative Routing Examples

The aggregate diagnostics above quantify *whether* the two-way baseline relies on the audio shortcut and *whether* TRIROUTE reroutes through visual evidence. To complement this with concrete cases, we mine FAVC FV-RA test clips on which the two-way baseline assigns very low fake probability ($P(\text{fake}) < 0.5$) while TRIROUTE assigns very high fake probability ($P(\text{fake}) > 0.5$), and rank by score gap. Of 3,007 FV-RA fake clips in the canonical FAVC test split, 1,048 satisfy this “two-way fails / TRIROUTE corrects” filter at seed 44, and the largest score gaps cluster tightly around 0.99. Figure ?? shows one representative case: the audio waveform looks like ordinary speech (RealAudio is genuine for FV-RA), and the lip-region frames show the manipulated visual stream that the two-way baseline ignores. The two-way baseline assigns $P(\text{fake}) \approx 0.01$ to this clip — it is “fooled” by the genuine audio — while TRIROUTE assigns $P(\text{fake}) \approx 1.00$. Five additional cases are shown in Figure ?? in the appendix, all drawn from the same top-gap pool and exhibiting the same pattern. Together with the head-ablation and input-perturbation evidence above, these cases illustrate the failure mode the three-way factorization is designed to prevent: when audio is real, a non-factorized detector has no incentive to use the visual stream, and confidently mislabels manipulated visual content as authentic.

6.4 Input-Level Routing Verification

The head-ablation analysis zeroes out internal model components, which could conflate routing with capacity effects. As a complementary and model-agnostic test, we perturb the *input features* on the FAVC FV-RA category and measure the resulting AUC change. This is a direct input-side intervention: if the trained head routes FV-RA decisions through the visual channel, full visual masking should cause a large drop; full audio masking, by contrast, should be neutral or beneficial (since FV-RA has real audio, masking it removes a potentially misleading signal).

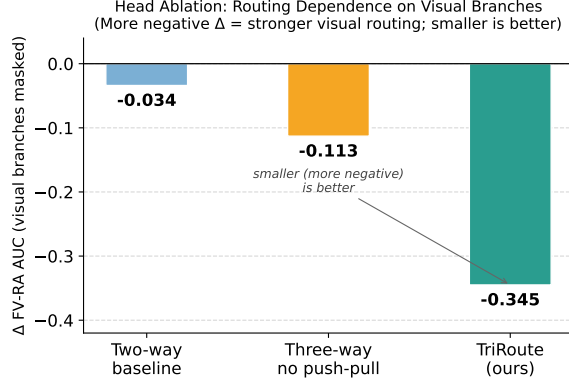


Figure 3: Head-ablation routing dependence on the visual branches, summarising Table ?? . Each bar is the magnitude of the FV-RA AUC drop when visual branches are masked. Only the factorized TRIROUTE model is meaningfully sensitive to removing visual components: the drop is roughly an order of magnitude larger than the two-way baseline (-0.345 vs. -0.034).

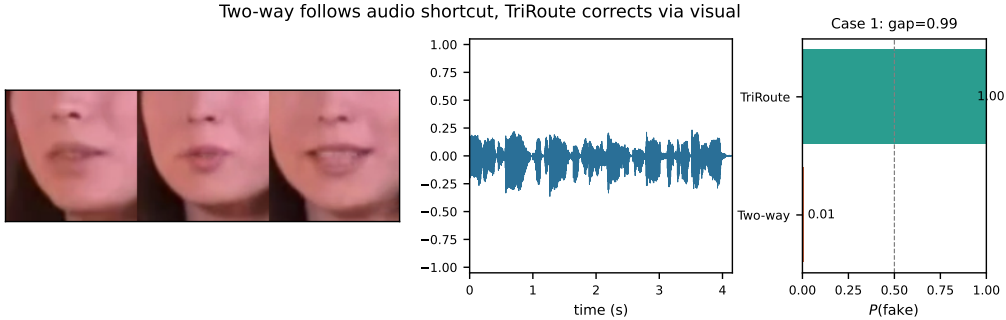


Figure 4: Representative FV-RA case where the two-way baseline misclassifies (follows the audio shortcut) and TRIROUTE corrects (routes through the visual stream). *Left*: three lip-region frames sampled across the clip, showing the manipulated visual stream. *Middle*: the audio waveform; this is genuine RealAudio, which a non-factorized detector reads as evidence of authenticity. *Right*: model fake-probabilities — the two-way baseline assigns $P(\text{fake}) \approx 0.01$, while TRIROUTE assigns $P(\text{fake}) \approx 1.00$. Five additional cases in Figure ??.

Table ?? reports mean FV-RA AUC over three independent seeds at full audio masking ($p = 1.0$) and full visual masking ($p = 1.0$). Two patterns stand out. First, the two-way baseline barely changes under either masking ($\Delta_{\text{audio}} = -0.021$, $\Delta_{\text{visual}} = -0.077$), consistent with a model near chance on FV-RA that is not strongly routing through either modality. Second, the latest TRIROUTE checkpoints show a strongly asymmetric profile: full visual masking lowers FV-RA from 0.838 to 0.461 on average ($\Delta = -0.377$), while full audio masking is approximately neutral ($\Delta = +0.005$). Three-way factorization without push-pull shows an intermediate profile ($\Delta_{\text{audio}} = +0.040$, $\Delta_{\text{visual}} = -0.137$), matching the intermediate FV-RA AUC in Table ??.

Progressive masking sweep (mechanism-level evidence). To go beyond the binary on/off intervention in Table ??, we sweep the mask probability $p \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0\}$ separately on the visual and audio channels. This continuous sweep is a stronger mechanism-level test than the binary intervention: it asks not only whether the model uses visual evidence, but whether its FV-RA response degrades smoothly and monotonically as that evidence is progressively removed. Figure ?? and Table ?? show that the curves are smooth and monotonic in the expected directions: visual masking lowers FV-RA AUC, audio masking raises it, and the magnitude of both effects grows from the two-way baseline through three-way (no push-pull) to TRIROUTE. The two-way baseline is nearly flat under visual masking ($\Delta = -0.037$) and only mildly responsive to audio masking ($+0.095$), consistent with weak visual routing. TRIROUTE shows the steepest visual-masking decline

Table 4: FV-RA AUC under full input masking on the FakeAVCeleb FV-RA category. Each row reports the mean over three independent seeds. Δ = masked AUC – baseline AUC. A positive Δ_{audio} indicates that removing the audio signal *helps* FV-RA detection, i.e., the model is actively exploiting the visual channel and suppressing misleading real-audio signal. Baseline values should be interpreted within this intervention protocol; the relative ordering matches the main table.

Model	Baseline	Δ_{audio} (full)	Δ_{visual} (full)	Relative visual sensitivity
Two-way baseline	0.546	−0.021	−0.077	1.0×
Three-way, no push-pull	0.616	+0.040	−0.137	1.8×
TriROUTE (ours)	0.838	+0.005	−0.377	4.9×

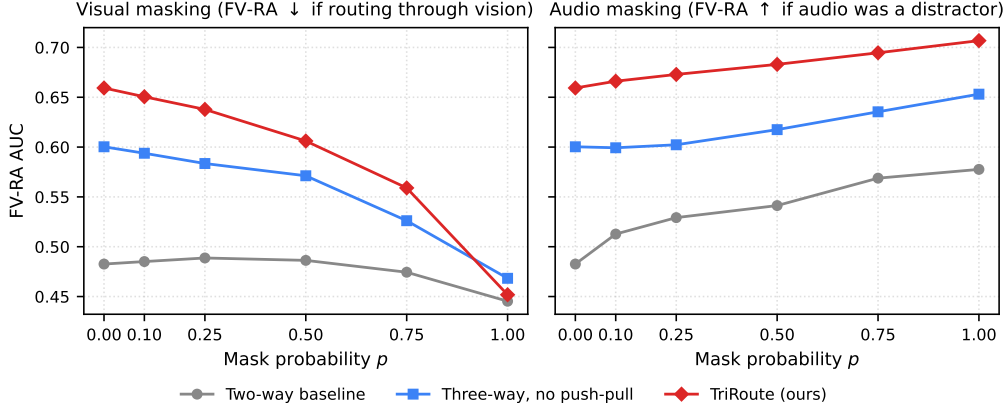


Figure 5: Progressive input masking on the FAVC FV-RA category. *Left*: masking the visual channel monotonically lowers FV-RA AUC, with the steepest drop for TriROUTE. *Right*: masking the audio channel monotonically raises FV-RA AUC, indicating the real-audio stream was acting as a distractor that the visual-routed detector benefits from suppressing. The two-way baseline is nearly flat under visual masking, consistent with weak visual routing.

368 ($\Delta = -0.208$) and a clear positive response to audio masking (+0.047), indicating that the detector
 369 has learned to exploit visual evidence and tolerate or actively suppress real-audio distractors.

370 6.5 Subspace Probe Analysis

371 To check that the structural separation actually concentrates transferable visual evidence into a
 372 specific component rather than redistributing it diffusely, we train light linear probes directly on
 373 individual subspaces and report two complementary scores: cross-domain (OOD) FV-RA AUC, and a
 374 domain-classification AUC that measures how strongly each subspace encodes source/target identity.
 375 A useful manipulation pathway should be predictive of FV-RA *and* relatively domain-blind.

376 Two patterns are visible in Table ?? and Figure ?. **First, the cross-domain signal in TriROUTE is**
 377 **concentrated in a strictly lower-dimensional subspace.** The 256-dim modality-unique factor u_v
 378 attains 0.924 OOD probe AUC, which is higher than both the 1024-dim fused two-way representation
 379 (0.705) and the 512-dim modality-mixed two-way visual factor (0.891). In other words, TriROUTE
 380 achieves *higher OOD performance despite lower dimensionality*: the structural separation isolates
 381 a more transferable manipulation pathway in u_v , rather than relying on a wider but more shortcut-
 382 contaminated representation.

383 **Second, this gain is mechanism-level evidence of routing rather than of domain invariance.** The
 384 domain-classification probe stays high on every subspace (0.967–0.998, see Table ?? right column),
 385 and TriROUTE’s u_v in particular still reaches 0.976 domain probe AUC. Domain identity is therefore
 386 not erased – it is just partly displaced *away* from the prediction-relevant subspace, consistent with the
 387 push-pull objective. We explicitly do not claim full domain invariance: u_v remains weakly domain-
 388 predictive, but the manipulation signal is now concentrated in a subspace where domain identity is the
 389 weaker explanation. Figure ?? provides a qualitative complement: t-SNE projections show that the

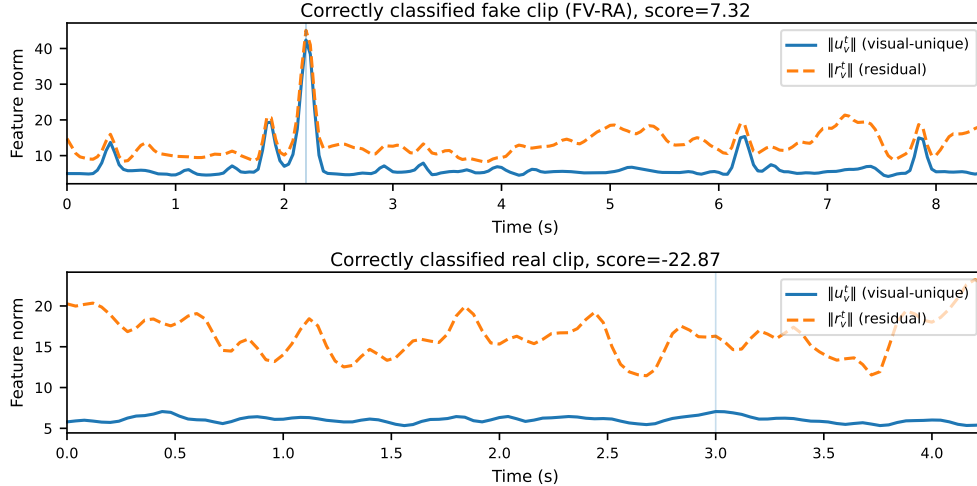


Figure 6: Per-frame routing map for two FAVC clips classified by TRIROUTE. We plot the time-varying feature norm of the visual-unique factor $\|u_v^t\|$ (solid blue) and the residual factor $\|r_v^t\|$ (dashed orange). *Top*: a correctly classified fake (FV-RA) clip — $\|u_v^t\|$ has sharp peaks aligned with manipulation events, with the residual remaining background-level. *Bottom*: a correctly classified real clip — $\|u_v^t\|$ stays low while the residual carries the bulk of the activation. The detector reads only u_v (and s_v) at the prediction head, so the asymmetry in $\|u_v^t\|$ between the two clips is what drives the score.

Table 5: Progressive input masking on FAVC FV-RA AUC. Top block: visual masking ($\Delta < 0$ indicates visual routing). Bottom block: audio masking ($\Delta > 0$ indicates audio was a distractor). Numbers correspond to Figure ??.

Model	$p=0.0$	$p=0.1$	$p=0.25$	$p=0.5$	$p=0.75$	$p=1.0$	$\Delta(0 \rightarrow 1)$
<i>Visual masking</i>							
Two-way baseline	0.483	0.485	0.489	0.486	0.474	0.445	-0.037
Three-way, no push-pull	0.600	0.594	0.584	0.571	0.526	0.468	-0.132
TRIROUTE (ours)	0.659	0.651	0.638	0.606	0.559	0.452	-0.208
<i>Audio masking</i>							
Two-way baseline	0.483	0.513	0.529	0.541	0.569	0.578	+0.095
Three-way, no push-pull	0.600	0.599	0.602	0.618	0.635	0.653	+0.053
TRIROUTE (ours)	0.659	0.666	0.673	0.683	0.695	0.707	+0.047

390 residual branch separates source/target datasets into distinct clusters, while the manipulation factor
391 separates real from fake regardless of dataset — exactly the routing the factorization is designed to
392 produce.

393 6.6 Corrected Artifact Control

394 A potential confound in AVIM-trained models is leading silence: fabricated clips sometimes begin
395 with audio-free frames that are absent in authentic samples. We therefore evaluate a leading-silence-
396 trimmed version that removes raw-waveform leading silence prior to AV-HuBERT feature extraction.
397 Table ?? reports the trimming-induced change $\Delta = \text{AUC}_{\text{trimmed}} - \text{AUC}_{\text{untrimmed}}$ per category.
398 The pattern is a stark asymmetry: RV-FA and FV-FA, the two audio-fake categories, drop by tens
399 of AUC points after trimming, while FV-RA stays flat across every model ($|\Delta_{\text{FV-RA}}| \leq 0.008$). If
400 the FV-RA gain were a leading-silence artifact, trimming would erase it. The fact that it does not
401 means that FV-RA stability rules out audio shortcut explanations for the visual-routing gain. The
402 trimming-induced drops are instead concentrated where they should be, in the audio-fake categories,
403 indicating that the artifact mainly inflates audio-driven scoring. The full pre-/post-trim AUCs that
404 produce these Δ values are reported in Table ?? in the appendix.

Table 6: Subspace probe analysis. Linear probes are trained on each subspace and evaluated for cross-domain FV-RA AUC and domain-classification AUC. The visual-unique factor u_v in TRIROUTE achieves the strongest cross-domain probe AUC despite having only 256 dimensions, while its domain probe is comparatively lower than the two-way fused representation.

Model	Subspace (Dim.)	OOD probe AUC	Domain probe AUC
Two-way	task branch Z_c (1024)	0.705	0.998
Two-way	visual factor v_c (512)	0.891	0.994
Three-way, no push-pull	shared visual s_v (256)	0.814	0.992
Three-way, no push-pull	unique visual u_v (256)	0.825	0.967
TRIROUTE (ours)	shared visual s_v (256)	0.726	0.989
TRIROUTE (ours)	unique visual u_v (256)	0.924	0.976

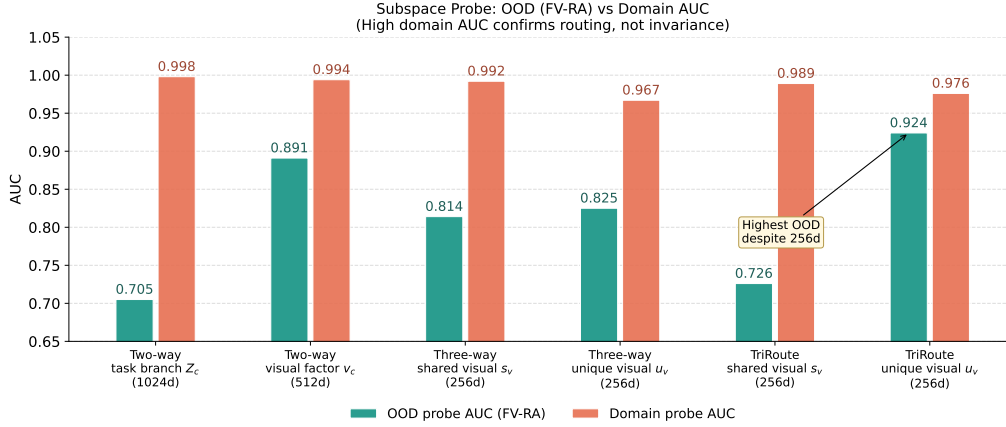


Figure 7: Subspace probe AUCs for OOD FV-RA prediction (teal) and source/target domain prediction (orange) across the six subspaces in Table ???. The visual-unique factor u_v in TRIROUTE achieves the strongest cross-domain probe AUC (0.924) despite its lower dimensionality (256d), while the domain probe stays high (0.976). High domain AUC across all subspaces indicates that the gain reflects routing rather than full domain invariance.

6.7 Domain Label Quality

A natural question is how much the quality of nuisance supervision matters. The speaker-group pseudo-label is not a target-domain label: it partitions source clips by speaker identity and is therefore closer to a weak speaker-group nuisance proxy than to true knowledge of the AVIM-to-FAVC domain gap. It does not directly encode codec differences, acquisition conditions, or any FAVC-specific artifact. Yet the latest TRIROUTE checkpoints raise FV-RA AUC from the two-way baseline of 0.437 to 0.838, a +0.401 gain without using any FakeAVCeleb data during training. We therefore interpret nuisance supervision as helpful but secondary to the representation split itself: even a weak source-side partition is enough to make the residual sink useful once the classifier is structurally prevented from reading it.

6.8 Comparison in AVH-Align Protocol

Table ?? replicates the structure of Table 2 in [?], reporting AUC and AP on the FakeAVCeleb 70/30 test split and AV-Deepfake1M validation set ($N = 2,000$), with and without leading-silence trimming. Baseline rows are reproduced from [?]. Our rows report the source-side nuisance-label variant averaged over three independent seeds. This table complements Table ???: Table ?? isolates the FV-RA visual-routing failure mode, while this table gives broader aggregate context against published audio-visual baselines.

We treat this comparison as supportive rather than central because it mixes source-supervised AVIM training with unsupervised speech-pretraining regimes. The most direct comparison is the supervised AVIM-trained AVH-Align row, since it uses the same transfer direction and compatible AV-HuBERT

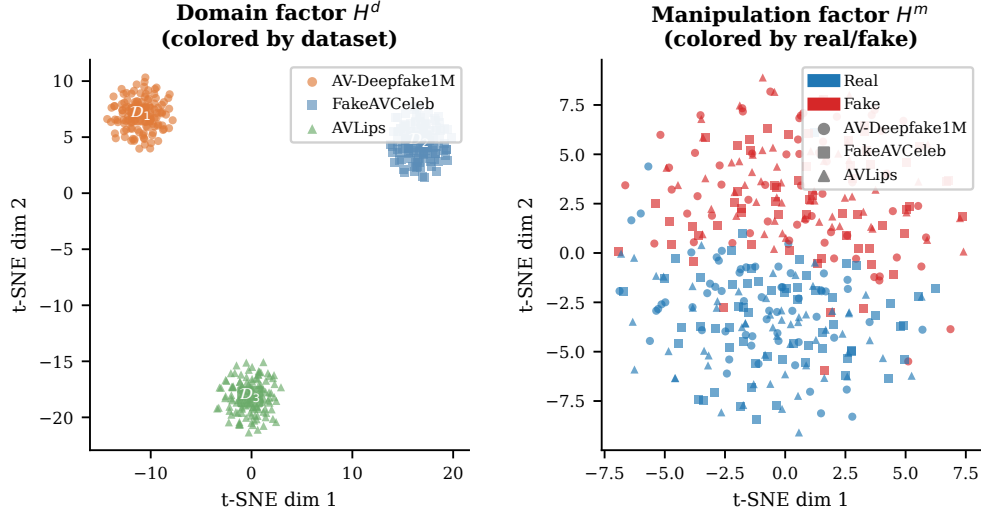


Figure 8: t-SNE projections of TRIROUTE’s residual and manipulation factors on held-out clips from AV-Deepfake1M, FakeAVCeleb, and AVLips. *Left*: the residual factor (“domain factor”) cleanly separates source and target datasets, confirming that domain-correlated nuisance is concentrated in the branch that is structurally excluded from the prediction head. *Right*: the manipulation factor separates real from fake regardless of dataset, indicating that the prediction-relevant subspace generalizes across domains.

Table 7: Trimming-induced change in AUC ($\Delta = \text{trimmed} - \text{untrimmed}$) on the FAVC categories, mean over three independent seeds. Audio-fake categories (RV-FA, FV-FA) drop substantially under trimming; FV-RA stays flat ($|\Delta| \leq 0.008$) across all three models. Pre-/post-trim raw AUCs are deferred to Table ?? in the appendix.

Model	Δ_{Overall}	$\Delta_{\text{RV-FA}}$	$\Delta_{\text{FV-RA (key)}}$	$\Delta_{\text{FV-FA}}$
Two-way baseline	−0.076	−0.038	−0.004	−0.141
Three-way, no push-pull	−0.050	−0.049	−0.008	−0.087
TRIROUTE (ours)	−0.056	−0.189	−0.005	−0.096

425 features. Under that comparison, after leading-silence trimming, TRIROUTE remains clearly stronger
426 on FAVC (86.6 vs. 70.8 AUC). The unsupervised VoxCeleb2-pretrained rows are included for
427 context rather than as matched baselines; their higher FAVC numbers indicate the remaining benefit
428 of broader speech pretraining rather than a failure of the routing analysis. Note that TRIROUTE
429 trained directly on FAVC reaches near-ceiling in-domain performance (AUC 99.6, lower block of
430 Table ??), indicating that the AV1M→FAVC gap is not a model-capacity limitation of TRIROUTE
431 but a manifestation of cross-dataset shift.

432 6.9 Supplementary External Support on AVLips

433 AVLips is a meaningful distribution shift relative to AV-Deepfake1M: it uses a different set of source
434 identities, capture conditions, and synthesis pipelines, and the t-SNE projection of the residual factor
435 in Figure ?? shows that AVLips clips form a cluster well-separated from both AV1M and FAVC.
436 To test whether the FV-RA-trained ordering carries over to this second external benchmark, we
437 evaluated all three model families on AVLips [?] and report results in Table ?. AVLips provides
438 only binary real/fake labels without modality-category annotations, so only full-set AUC and AP can
439 be reported. The same ordering is preserved: the two-way baseline reaches 0.533 AUC, three-way
440 factorization without domain push-pull reaches 0.588, and TRIROUTE reaches 0.789; all values are
441 means over three independent seeds. This result is directionally consistent with the FakeAVCeleb
442 benchmark and suggests that the routing-based improvement is not isolated to a single target dataset.
443 We nevertheless treat AVLips as supportive rather than central evidence, because AVLips does not
444 provide modality-category labels and therefore cannot replicate the FV-RA routing diagnostic.

Table 8: Comparison with ?] Table 2 baselines on FakeAVCeleb (FAVC) 70/30 test split and AV-Deepfake1M (AV1M) validation set. Values are percentages. Trim:✓ denotes leading-silence-trimmed evaluation. Baseline rows are reproduced from ?]; all our rows report the mean over three independent seeds. Bold highlights the best source-supervised AV1M-trained result within our matched comparison block, not the unsupervised VoxCeleb2-pretrained rows. FAVC-trained rows additionally show the reverse-direction transfer; FAVC Trim:✓ columns are identical to Trim:× for FAVC-trained models because leading-silence trimming targets AV1M-synthesized clips.

Method	Mod.	Train type	Train data	Metric: AUC				Metric: AP			
				FakeAVCeleb		AV-Deepfake1M		FakeAVCeleb		AV-Deepfake1M	
				Trim:×	Trim:✓	Trim:×	Trim:✓	Trim:×	Trim:✓	Trim:×	Trim:✓
AVH-Align/sup [?]	AV	sup.	FAVC	99.2	99.2	69.0	63.6	100.0	100.0	84.1	82.2
AVH-Align/sup [?]	AV	sup.	AV1M	77.5	70.8	100.0	83.1	99.4	99.1	100.0	94.4
AVAD [?]	AV	unsup.	LRS	84.5	84.7	54.3	54.3	99.5	99.5	76.3	76.3
SpeechForensics [?]	AV	unsup.	VoxCeleb2	98.8	98.8	68.8	68.2	100.0	100.0	83.7	83.5
AVH-Align [?]	AV	unsup.	VoxCeleb2	94.6	94.6	85.9	83.5	99.8	99.8	94.3	93.5
Two-way baseline (ours)	AV	sup.	AV1M	73.9	69.7	99.8	74.7	99.3	99.0	99.9	90.1
Three-way, no push-pull (ours)	AV	sup.	AV1M	82.7	78.0	99.9	84.6	99.6	99.4	100.0	94.7
TriROUTE (ours)	AV	sup.	AV1M	92.2	86.6	99.7	84.8	99.8	99.6	99.9	94.1
Two-way baseline (ours)	AV	sup.	FAVC	99.5	99.5	67.0	63.4	100.0	100.0	81.2	79.8
Three-way, no push-pull (ours)	AV	sup.	FAVC	99.7	99.7	68.3	65.1	100.0	100.0	82.2	80.4
TriROUTE (ours)	AV	sup.	FAVC	99.6	99.6	74.7	67.7	100.0	100.0	86.4	82.2

Table 9: AVLips cross-dataset evaluation ($N = 7,602$). Values are mean AUC/AP averaged over three independent seeds. AVLips provides only binary real/fake labels; per-modality-category breakdown is not available. Reported as supplementary directional evidence.

Model	AUC	AP
Two-way baseline	0.533	0.598
Three-way, no domain push-pull	0.588	0.648
TriROUTE (ours)	0.789	0.784

Perturbation companion on AVLips. To test whether the visual-routing pattern observed on FAVC also holds on this second target dataset, we run the same single-channel masking intervention on AVLips at the seed-44 checkpoint of each model: full visual masking ($p = 1.0$) and full audio masking ($p = 1.0$), reusing the masking definition from Table ?? . Table ?? reports the resulting changes in AUC. The pattern matches FAVC: the two-way baseline is essentially insensitive to either masking ($\Delta_{\text{vis}} = +0.014$, $\Delta_{\text{aud}} = -0.035$), consistent with a model that is near chance on AVLips and not strongly routing through either modality. TriROUTE, in contrast, shows a strongly asymmetric profile — full visual masking lowers AVLips AUC from 0.800 to 0.475 ($\Delta = -0.324$), while full audio masking is approximately neutral ($\Delta = -0.003$). Three-way factorization without push-pull again sits in between ($\Delta_{\text{vis}} = -0.048$, $\Delta_{\text{aud}} = +0.048$). The visual-routing asymmetry, in other words, replicates on AVLips even though AVLips does not expose per-category labels: the same intervention that exposes the routing on FAVC FV-RA also exposes it on the AVLips full set.

7 Limitations and Broader Impact

Limitations. TriROUTE is still anchored to speech-related evidence, so it may be less effective on deepfakes with minimal speaking motion or on clips where the audio track is absent, replaced, or intentionally desynchronized for benign reasons. The main category-level gain is also concentrated in FV-RA, because RV-FA and FV-FA are already near ceiling in the AV1M→FAVC protocol; this makes FV-RA the most diagnostic setting for visual routing, but it also limits how broadly the category-level result can be interpreted. The external supervised comparison is currently limited by the scarcity of published methods trained on AV1M and evaluated on FAVC under matched category annotations. Retraining additional baselines such as AVAD, SpeechForensics, and AVFF under the same AV1M→FAVC protocol would strengthen the empirical comparison.

More specifically, our baseline block should be read as a mechanism-oriented comparison rather than an exhaustive leaderboard. We compare against the closest published compatible supervised bridge (AVH-Align/sup) and against internal controls designed to isolate routing effects, but we do not retrain a broader external baseline suite under the exact AV1M→FAVC category protocol. Likewise,

Table 10: AVLips masking intervention at seed 44. Baseline AUC is on clean inputs. Δ_{vis} is the change after applying full visual masking; Δ_{aud} is the change after applying full audio masking (both $p = 1.0$, masking definition matches Table ??). The asymmetric drop under visual masking and neutral response under audio masking that we observed on FAVC FV-RA also appears on AVLips: only TRIROUTE shows a large $|\Delta_{\text{vis}}|$, indicating it is the only model routing predictions through the visual stream on this target dataset as well. Reported at single seed 44 for the perturbation eval (the baseline column in Table ?? is the multi-seed mean).

Model	Baseline AUC	Δ_{vis} (full)	Δ_{aud} (full)
Two-way baseline	0.507	+0.014	−0.035
Three-way, no push-pull	0.595	−0.048	+0.048
TRIROUTE (ours)	0.800	−0.324	−0.003

while DANN and CORAL provide two complementary global-alignment controls, we report them under matched default settings rather than a full GRL hyperparameter sweep; a stronger baseline audit would include a broader α sensitivity study and possibly alternative adversarial schedules.

The factorization introduces additional optimization complexity and benefits from some nuisance partition or domain contrast during training. As shown in Section ??, the speaker-group pseudo-label is an extremely weak proxy for the true domain gap, yet the factorization improves FV-RA substantially. Whether stronger or multi-source domain contrast could yield further gains remains an open question. Benchmark breadth is also still limited: FakeAVCeleb is the only fully analyzed target with category-level routing diagnostics in the latest result set. Finally, our routing evidence is mechanistic but not a formal causal proof: branch ablations and input masking both point to heavier use of the visual-unique factor, but future falsification tests should vary the capacity of u_v and the residual branch directly.

Broader impact. Cross-domain deepfake detection can support content authentication, media forensics, and platform integrity. At the same time, any detector can be misused for surveillance or overconfident moderation, especially if its uncertainty is not surfaced. We therefore recommend reporting calibration, failure modes, and domain-shift sensitivity, and releasing models with clear usage restrictions and dataset documentation.

8 Conclusion

We proposed TRIROUTE, a factorized audio-visual deepfake detector that separates shared content, modality-unique manipulation evidence, and residual domain-sensitive signal so that the prediction head is discouraged from solving the task through shortcut-heavy branches. Across the AV1M→FAVC cross-domain benchmark, the main result is a substantial improvement on FV-RA, the diagnostic category that forces the detector to rely on visual manipulation evidence: global DANN and CORAL controls on a coarser two-way representation remain near chance (0.496/0.955 and 0.521/0.956), while the latest TRIROUTE checkpoints reach 0.838 FV-RA AUC using only source-side nuisance labels. Head ablation and input-level perturbation both indicate heavier routing dependence on the visual-unique channel. Together, these analyses support a routing-based interpretation of the cross-domain gain rather than a claim of full domain invariance. For cross-domain forensics, the central question is therefore less how to make every feature domain-invariant, and more how to structurally prevent the detector from using shortcut features it can otherwise see. On FV-RA specifically, TRIROUTE moves the closest AV1M-supervised external baseline (AVH-Align/sup, 0.519 AUC) to 0.838 AUC in the latest run, shifting performance from near-chance transfer to a practically usable cross-domain detector on the visual-routing stress category. We therefore view the present evidence as strongest for the routing mechanism itself under a deliberately diagnostic cross-domain setting, rather than as a final statement about universal superiority across datasets, baselines, or adaptation schedules.

A Full Leading-Silence Trim Table

Table ?? in the main text reports trimming-induced changes (Δ) only. Table ?? below reports the full pre-trim and post-trim AUCs that produce those Δ values, including all four FAVC categories. The asymmetry highlighted in the main text is visible directly here: the audio-fake categories (RV-FA, FV-FA) lose substantial AUC after trimming, while FV-RA stays within 0.008 AUC of its untrimmed value across all three models.

Table 11: Full leading-silence trim AUCs on FakeAVCeleb categories, mean over three independent seeds. “Untrimmed” is the reference condition reported in the main results; “Trimmed” removes raw-waveform leading silence prior to AV-HuBERT feature extraction. $\Delta = \text{Trimmed} - \text{Untrimmed}$. FV-RA is highlighted; it is the visual-routing diagnostic and stays flat across all three models, while RV-FA and FV-FA (audio-fake categories) drop substantially.

Model	Condition	Overall	RV-FA	FV-RA	FV-FA
Two-way baseline	Untrimmed AUC	0.777	1.000	0.512	1.000
Two-way baseline	Trimmed AUC	0.701	0.962	0.508	0.859
Two-way baseline	Δ	−0.076	−0.038	−0.004	−0.141
Three-way, no push-pull	Untrimmed AUC	0.831	0.999	0.634	0.996
Three-way, no push-pull	Trimmed AUC	0.781	0.950	0.626	0.909
Three-way, no push-pull	Δ	−0.050	−0.049	−0.008	−0.087
TriROUTE (ours)	Untrimmed AUC	0.922	0.999	0.838	0.994
TriROUTE (ours)	Trimmed AUC	0.866	0.810	0.833	0.898
TriROUTE (ours)	Δ	−0.056	−0.189	−0.005	−0.096

B Additional Qualitative Routing Examples

Figure ?? reports five additional FV-RA test clips drawn from the same “two-way fails / TriROUTE corrects” pool described in Section ?. The clips are the next-five highest-gap cases at seed 44 after the representative case in Figure ?. The pattern is consistent across cases: the audio waveform appears as ordinary speech, the lip-region frames carry the manipulation evidence, the two-way baseline assigns $P(\text{fake}) \approx 0.01$ in every case, and TriROUTE assigns $P(\text{fake}) \approx 0.99$ or higher. Combined, the six cases in Figures ?? and ?? illustrate the audio-shortcut failure mode that motivates the three-way factorization, complementing the aggregate evidence in Tables ??, ??, and ??.

Table ?? summarises the six cases and the expected detector behavior in each. Every case is FV-RA (real audio paired with a manipulated lip-synced video stream), so the expected ground-truth-aligned response is $P(\text{fake})$ close to 1. The two-way baseline returns $P(\text{fake}) \approx 0.01$ in every case (audio shortcut), while TriROUTE returns $P(\text{fake}) \gtrsim 0.99$ via the visual-unique branch. We retain the full per-case visualisation in Figures ?? and ?? so that lip-region content and audio waveforms can be inspected directly; the table is a compact companion view.

Table 12: Compact summary of qualitative routing examples (FAVC FV-RA, seed 44, top-gap cases). For all six cases the ground-truth label is fake (manipulated video) and the audio is genuine, so the expected $P(\text{fake})$ is close to 1. Two-way assigns near-zero in every case (audio shortcut); TriROUTE correctly assigns near-one. The score gap is $P_{\text{TriROUTE}}(\text{fake}) - P_{\text{Two-way}}(\text{fake})$. “Synth.” refers to the FAVC visual manipulation pipeline encoded in the clip filename (Wav2Lip-style for *wavtolip* cases; remaining cases use clip-id codes that resolve to the FAVC face-reenactment pipelines).

#	Demographic	Synth.	Clip ID	Truth	Expect	$P_{\text{Two-way}}$	P_{TriROUTE}
1	Caucasian (Eu.), F	wavtolip	00092_id00271	fake	→ 1	0.01	1.00
2	Caucasian (Eu.), F	wavtolip	00043_id00633	fake	→ 1	0.01	1.00
3	Asian (S.), M	face-reenact	00021_id07108	fake	→ 1	0.01	1.00
4	Caucasian (Am.), M	wavtolip	00243_id00062	fake	→ 1	0.01	1.00
5	Asian (S.), F	wavtolip	00069_id07689	fake	→ 1	0.01	1.00
6	Caucasian (Eu.), F	wavtolip	00092_id00188	fake	→ 1	0.01	1.00

527 The six cases span four demographic strata (Caucasian-European F, Caucasian-American M, Asian-
528 South M/F, with a face-reenact pipeline case in row 3) and two synthesis families (lip-sync *wavtolip*
529 and a non-lip-sync face-reenactment case), so the failure mode is not concentrated on a single
530 subgroup or pipeline. In every row, the audio waveform is genuine (FV-RA condition) and the
531 only signal that can carry the fake label is visual; the two-way baseline does not use it, while the
532 TriROUTE task head reads visual-unique u_v and assigns the correct label.

Additional FV-RA cases where two-way fails / TriRoute corrects

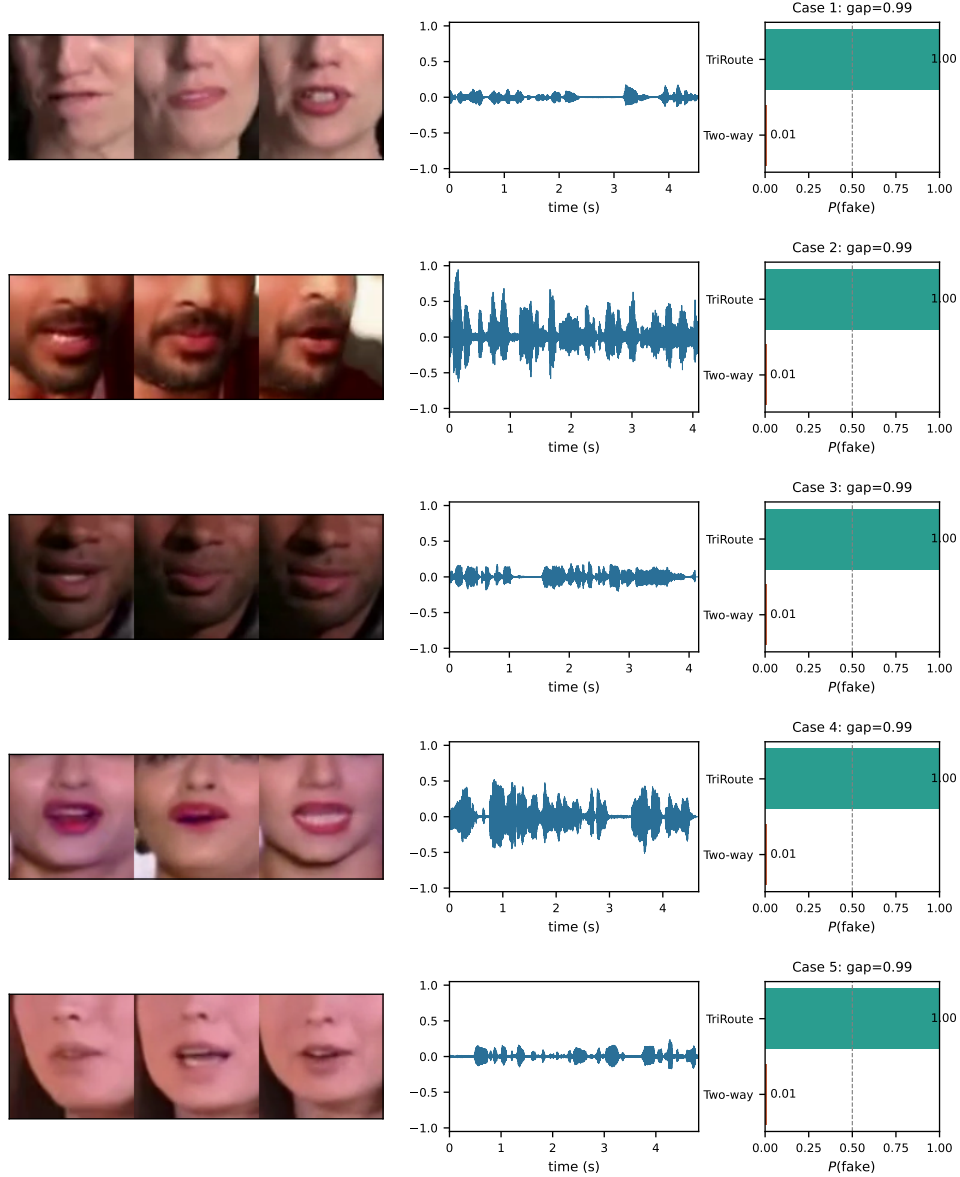


Figure 9: Five additional FV-RA cases (highest score gap after the representative case in Figure ??). For each row, *left*: three lip-region frames; *middle*: audio waveform; *right*: model fake-probabilities. In every case the two-way baseline assigns $P(\text{fake}) \approx 0.01$ (audio-shortcut failure) and TriROUTE assigns $P(\text{fake}) \approx 0.99$.

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